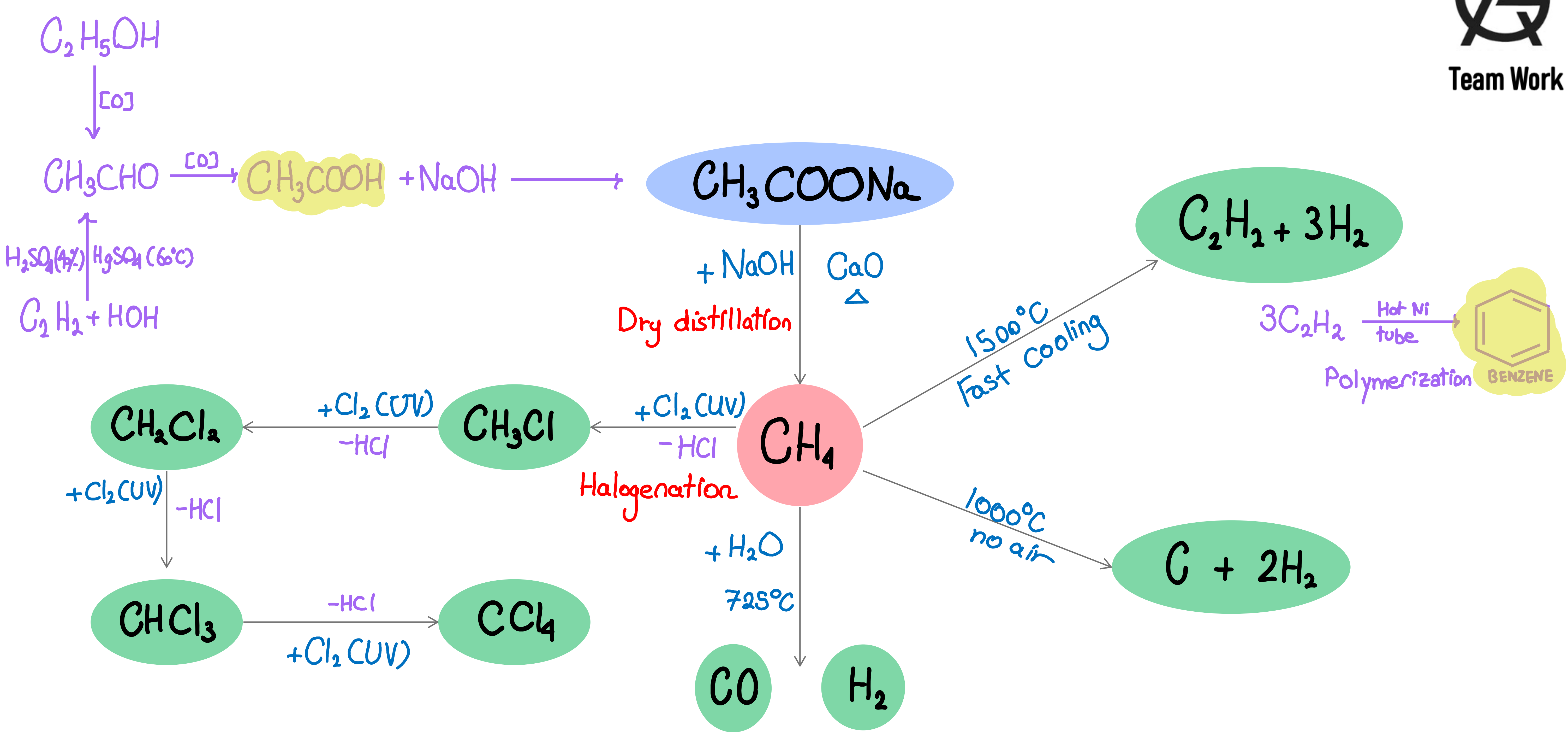
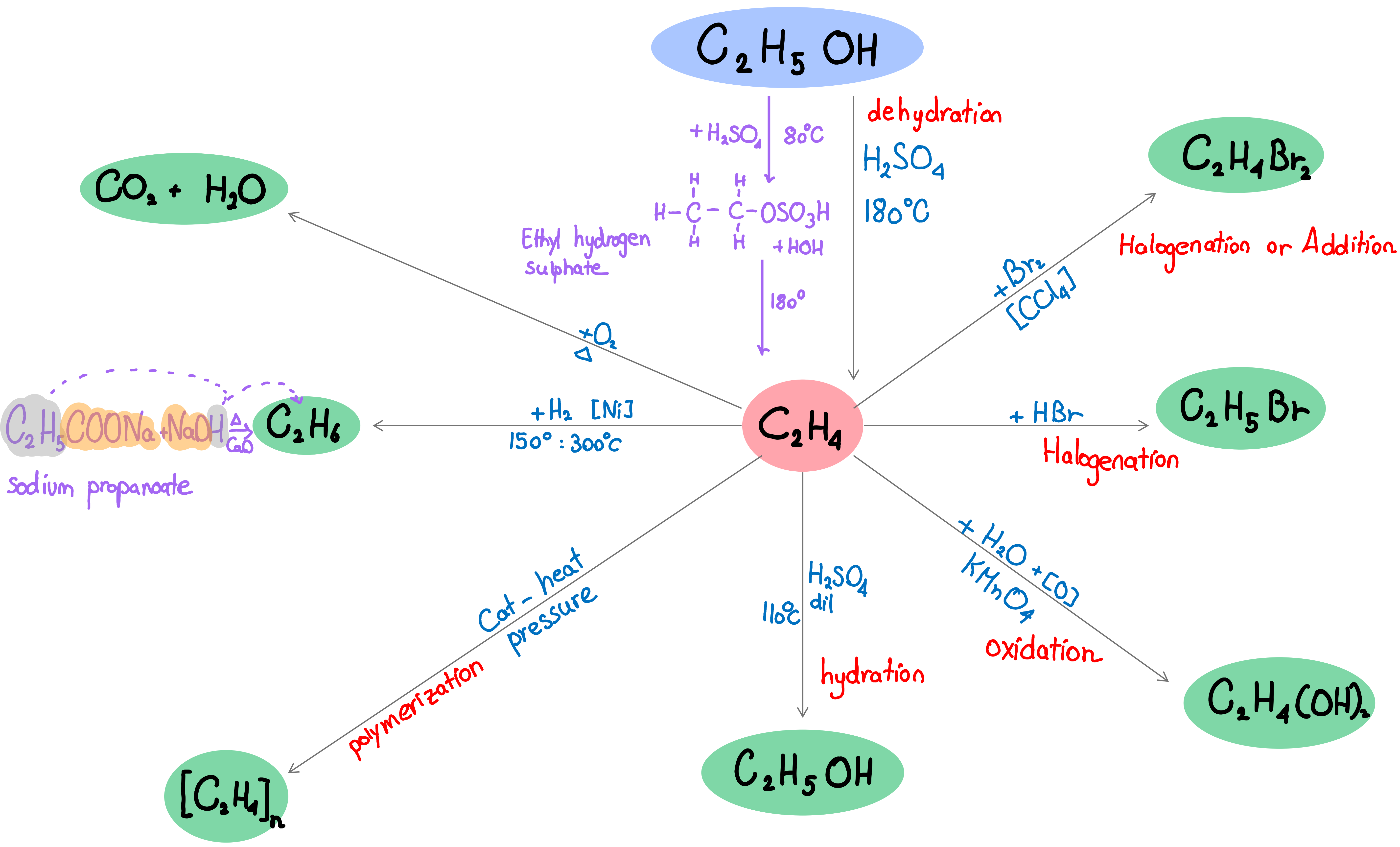
Methane Diagram

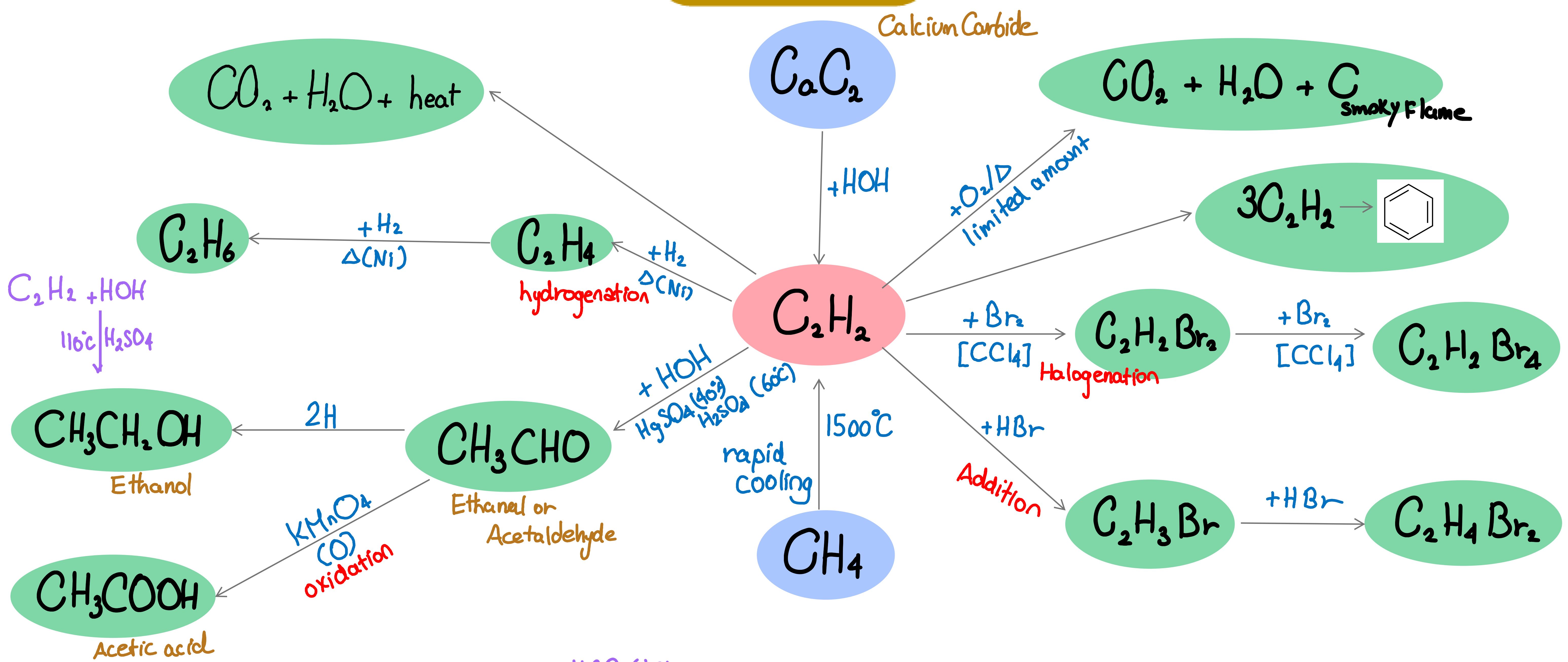
Team Work



Ethene Diagram

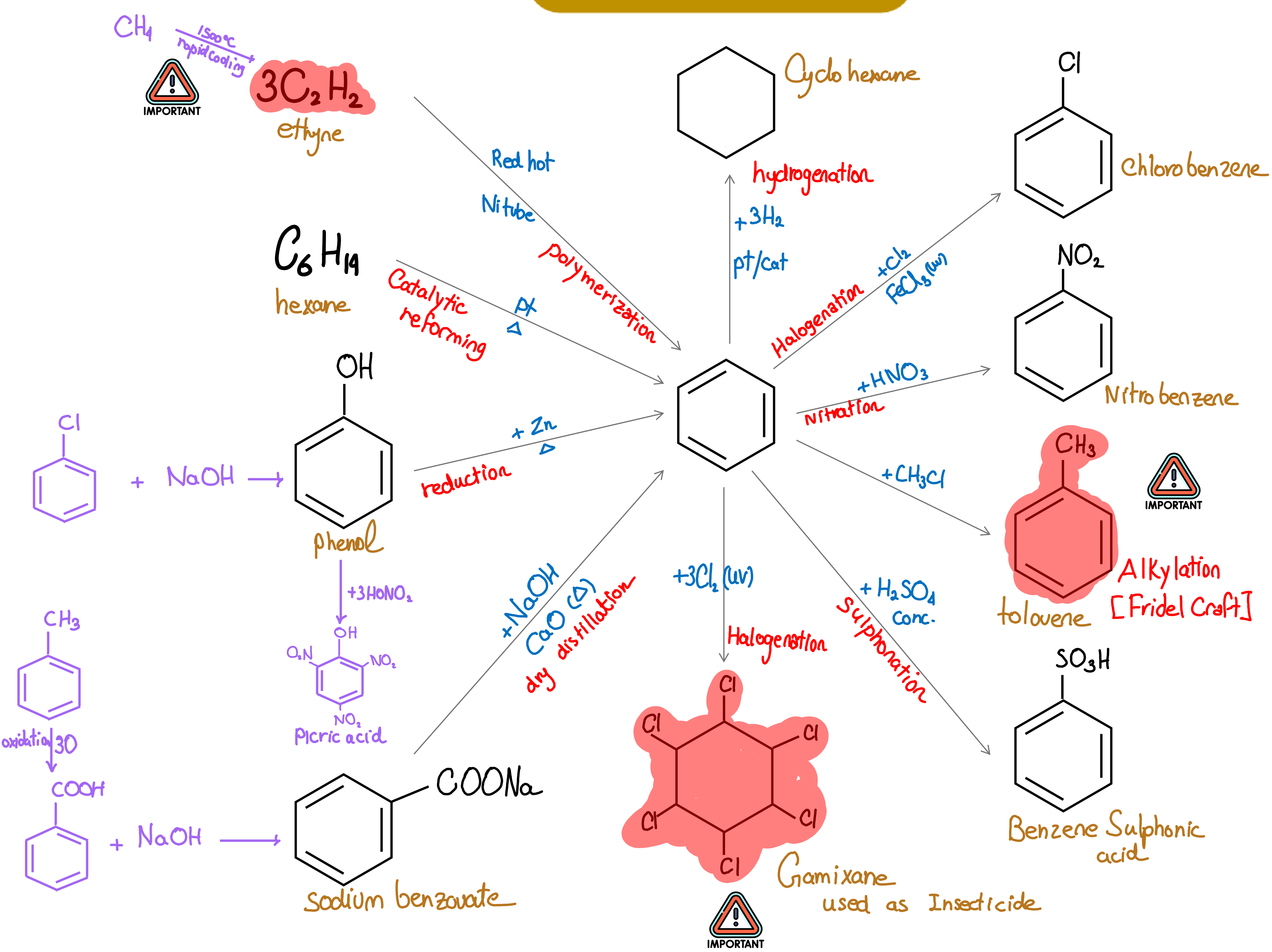


Ethyne Diagram



IMPORTANT **Note:** $HC \equiv CH + HOH \xrightarrow[H_2SO_4(60\%)]{H_2SO_4(40\%)}$ **Hydration** $H_2C = CH(OH)$ Vinyl alcohol $\xrightarrow{\text{rearrangment}}$ $H_3C-C(=O)-H$ ----- CH_3CHO

Aromatic Benzene Diagram



Important Diagrams

$C_2H_5OH \xrightarrow{[O]} CH_3CHO \xrightarrow{[O]} CH_3COOH \xrightarrow{+NaOH} CH_3COONa$ sodium Acetate

ethanol, acetaldehyde, aceticacid

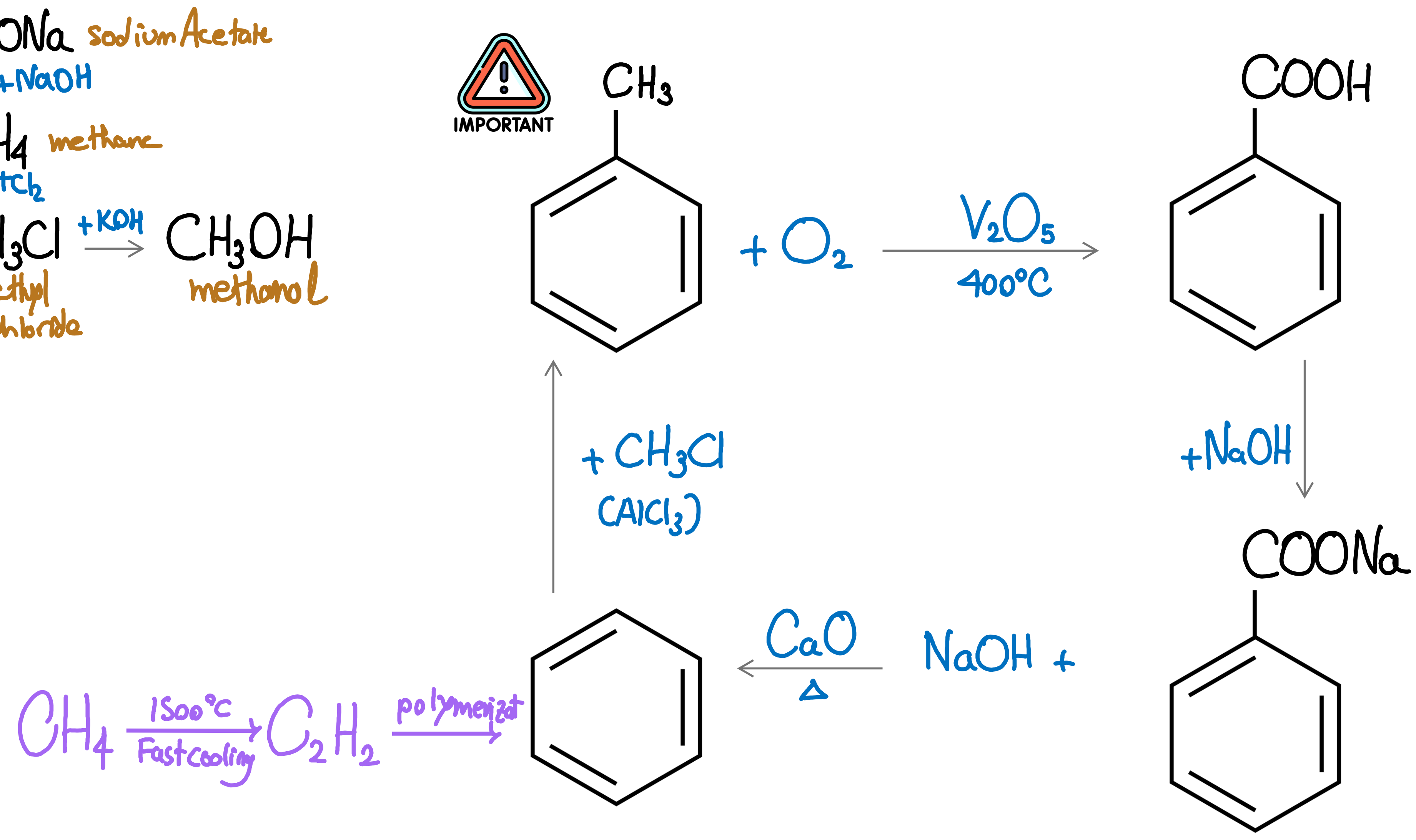
$CH_3COONa \xrightarrow{+NaOH} CH_4$ methane

$CH_4 \xrightarrow{+Cl_2} CH_3Cl \xrightarrow{+KOH} CH_3OH$

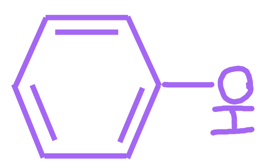
methyl chloride, methanol

IMPORTANT

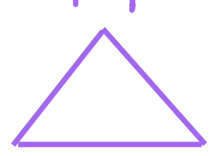
$CH_4 \xrightarrow[\text{Fast Cooling}]{1500^\circ C} C_2H_2 \xrightarrow[H_2SO_4/H_2SO_4]{+H_2O} CH_3CHO \xrightarrow[\text{acidified } KMnO_4]{[O]} CH_3COOH \xrightarrow{+NaOH} CH_3COONa \xrightarrow[\Delta]{+NaOH,\ CaO} CH_4$



Functional Groups

ClASS	FUNCTIONAl GROUP	EXAMPLE
Alcohols	—OH hydroxyl	CH_3OH methanol
Phenols	—OH hydroxyl + Benzene ring	 phenol
ethers	R—O—R	$C_2H_5-O-C_2H_5$ diethyl ether
Aldehyde	$R-\overset{O}{\overset{\Vert}{C}}-H$	$CH_3\overset{O}{\overset{\Vert}{C}}-H$ or CH_3CHO acetaldehyde
Ketones	$R-\overset{O}{\overset{\Vert}{C}}-R$	$CH_3-\overset{O}{\overset{\Vert}{C}}-CH_3$ Acetone
Acids	$R-\overset{O}{\overset{\Vert}{C}}-OH$ $-COOH$	CH_3COOH acetic acid
esters	$R-\overset{O}{\overset{\Vert}{C}}-O-R$ $-COO-$	$CH_3-\overset{O}{\overset{\Vert}{C}}-O-C_2H_5$ ethyl acetate ester

Isomers According to General Formula

C_nH_{2n}	$C_nH_{2n}O$	$C_nH_{2n+2}O$	$C_nH_{2n}O_2$
1 Alkenes 2 Cyclo alkanes Example: $CH_3-CH=CH_2$ propene  Cyclopropane	1 Aldehydes 2 Ketones Ex: $CH_3-CH_2-\overset{O}{\overset{\Vert}{C}}-H$ propanal $CH_3-\overset{O}{\overset{\Vert}{C}}-CH_3$ Acetone	1 Monohydric Alcohol 2 Ethers Ex: C_2H_5OH ethanol CH_3-O-CH_3 dimethyl ether	1 Mono Carboxylic acids 2 Esters Ex: CH_3COOH acetic acid $H-\overset{O}{\overset{\Vert}{C}}-O-CH_3$ methyl methanoate ester

Directing Groups

DIRECTS TOWARDS ORTHO & PARA		DIRECTS TOWARDS META	
1 Alkyl (CH_3)	2 Hydroxyl ($-OH$)	1 Aldehyde group ($-CHO$)	2 Ketonic group ($-\overset{O}{\overset{\|\|}{C}}-$)
3 Amino ($-NH_2$)	4 Halogen (I, Br, Cl, F)	3 Carboxylic ($-COOH$)	4 Nitro (NO_2)

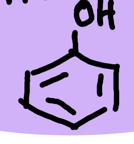
Oxidation and Reduction of functional Groups

FUNCTIONAL GROUP	OXIDATION	REDUCTION
Aldehyde group $-CHO$	Can be oxidized forming acid $-COOH$	Can be reduced forming primary Alcoh $-CH_2-OH$
Ketone group $-\overset{O}{\overset{\|\|}{C}}-$	Can't be oxidized	Can be reduced forming Secondary Alcohol $-\overset{\|}{C}HOH$
Alkenes $-C=C-$	Can be oxidized forming dihydric alcohol $-\underset{OH}{\underset{\|}{C}}-\underset{OH}{\underset{\|}{C}}-$	----
Carboxyl $COOH$	Can't be oxidized	Can be reduced partial: $-CHO$ Full: $-CH_2OH$ Forming Aldehyde or primary alcohol
primary Alcohol CH_2-OH	Can be oxidized partial: $-CHO$ Full: $-COOH$ Forming Aldehyde or Acid	Can't be reduced
secondary Alcohol $-\overset{\|}{C}H-OH$	Can be oxidized $-\overset{O}{\overset{\|\|}{C}}-$ Forming ketone	Can't be reduced
Tertiary Alcohol $-\overset{\|}{\underset{\|}{C}}-OH$	Can't be oxidized	Can't be reduced

$$\underset{CH_3CH_2OH}{\text{primary Alcohol}} \underset{H_2}{\overset{[O]\ \text{oxidation}}{\rightleftharpoons}} \underset{CH_3CHO}{\text{Aldehyde}} \underset{H_2\ \text{reduction}}{\overset{[O]}{\rightleftharpoons}} \underset{CH_3COOH}{\text{Acid}}$$

$$\underset{CH_3\overset{CH_3}{\overset{|}{C}}HOH}{\text{secondary Alcohol}} \underset{H_2}{\overset{[O]}{\rightleftharpoons}} \underset{CH_3-\overset{O}{\overset{||}{C}}-CH_3}{\text{Ketone}}$$

tertiary Alcohol $CH_3-\overset{CH_3}{\overset{|}{\underset{CH_3}{\underset{|}{C}}}}-OH$ can't be oxidized

phenols OH  Can't be oxidized

Team Work

Structural & Molecular Formulas of Important Aliphatic Hydrocarbons Derivatives

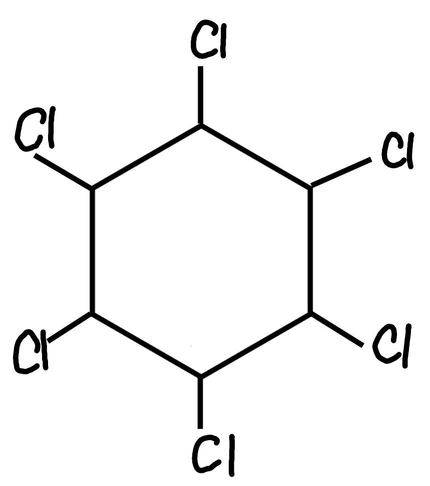
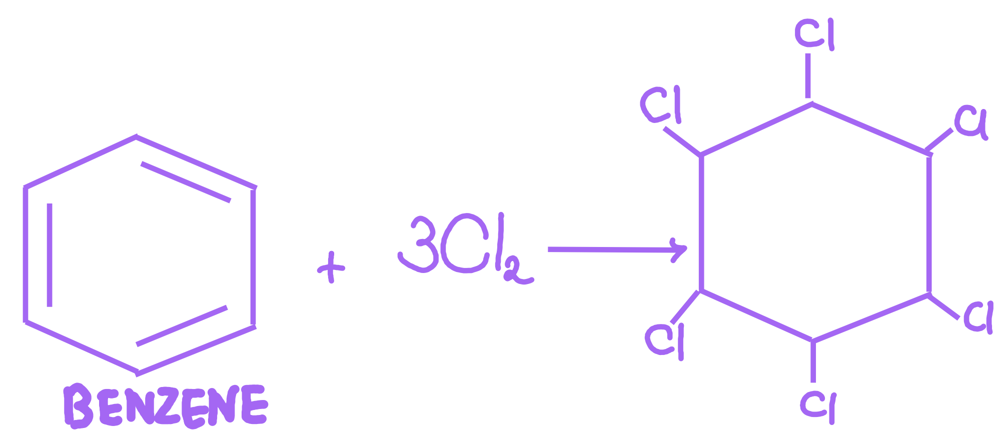
COMPOUND NAME	STRUCTURAL FORMULA	MOLECULAR FORMULA	PREPERATION
Methanol	CH_3-OH	CH_4O	$CH_3Cl + NaOH \longrightarrow CH_3OH + NaCl$ Alkaline hydrolysis
Ethanol	C_2H_5OH	C_2H_6O	① $C_2H_5Cl + NaOH \longrightarrow C_2H_5OH + NaCl$ Alkaline hydrolysis ② $\underset{\text{ethene}}{C_2H_4} + HOH \xrightarrow[H_2SO_4]{conc.} C_2H_5OH$ Catalytic hydration
propanol primary Alcohol	C_3H_7OH	C_3H_8O	① $C_3H_7Cl + NaOH \longrightarrow C_3H_7OH + NaCl$ Alkaline hydrolysis
2-propanol Secondary Alcohol	$\overset{1}{H_3C}-\overset{2}{CH}(OH)-\overset{3}{CH_3}$	C_3H_8O	① $CH_3-CH(Br)-CH_3 + NaOH \longrightarrow CH_3-CH(OH)-CH_3$ $+NaCl$ Alkaline hydrolysis ② $H_3C-CH=CH_2 + HOH \xrightarrow[H_2SO_4]{conc.} H_3C-CH(OH)-CH_3$ Catalytic hydration 2-propanol
2-methyl,2-propanol Tertiary Alcohol	$\overset{1}{H_3C}-\overset{2}{C}(CH_3)(OH)-\overset{3}{CH_3}$	$C_4H_{10}O$	① $CH_3-C(CH_3)(Br)-CH_3 + NaOH \longrightarrow CH_3-C(CH_3)(OH)-CH_3$ $+NaCl$ Alkaline hydrolysis ② $H_3C-C(CH_3)=CH_2 + HOH \xrightarrow[H_2SO_4]{conc.} H_3C-C(CH_3)(OH)-CH_3$ Catalytic hydration 2-methyl 2-propanol
IMPORTANT Ethylene glycol Dihydric Alcohol	$H_2C(OH)-CH_2(OH)$	$C_2H_6O_2$	$\underset{\text{ethene}}{H_2C=CH_2} + H_2O_2 \xrightarrow[\text{[O]}]{KMnO_4} H_2C(OH)-CH_2(OH)$ 2OH C_2H_4 Bayer's reaction
Glycerol Trihydric Alcohol	$\underset{\text{primary}}{H_2C(OH)}-\underset{\text{secondary}}{CH(OH)}-\underset{\text{primary}}{CH_2(OH)}$	$C_3H_8O_3$	

Alkaline hydrolysis of Alkyl halide can produce primary Secondary or tertiary Alcohol

Catalytic hydration of alkenes can produce only Secondary or tertiary alcohol exept for ethene produces ethanol

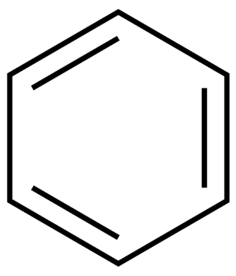
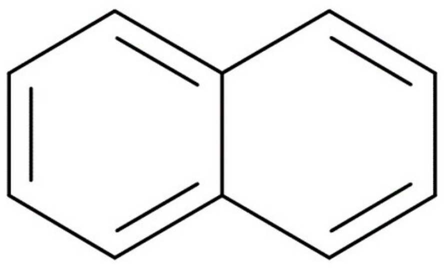
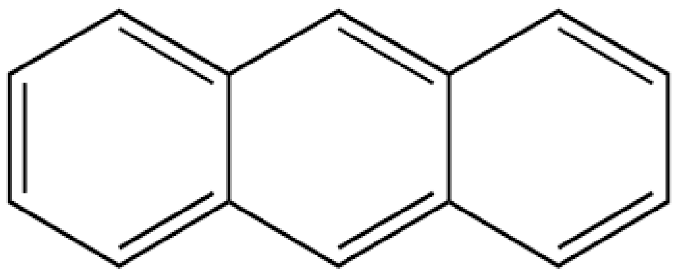
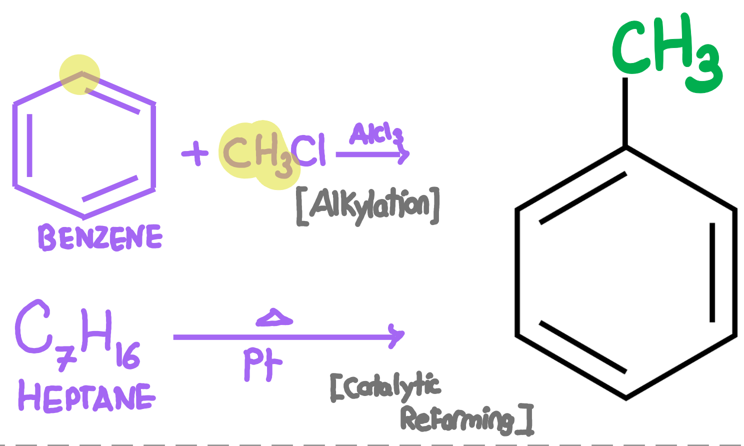
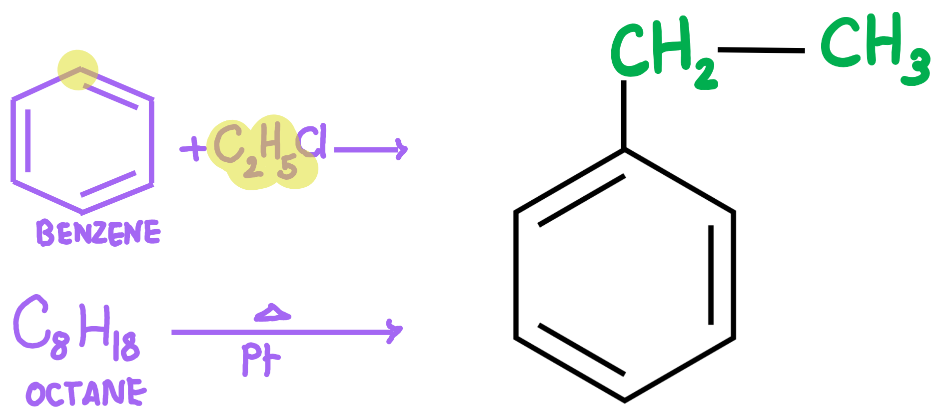
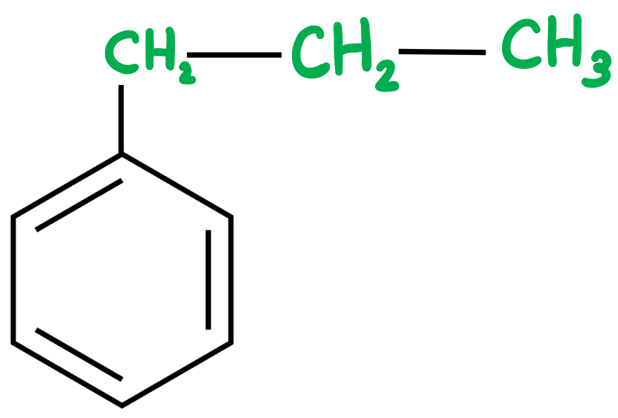
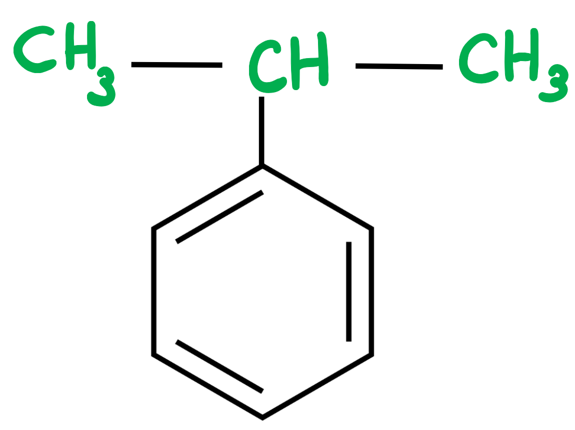
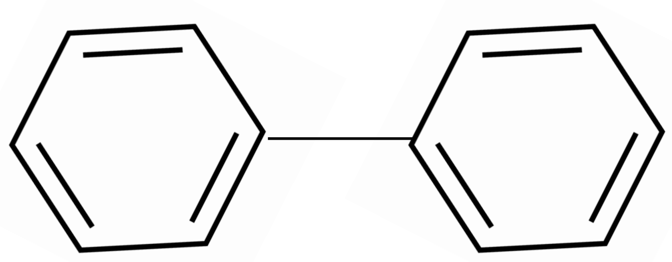
Team Work

COMPOUND NAME	STRUCTURAL FORMULA	MOLECULAR FORMULA	PREPERATION
Sorbitol poly hydric Alcohol	$CH_2(OH)-CH(OH)-CH(OH)-CH(OH)-CH(OH)-CH_2(OH)$ 6 - OH groups	$C_6H_{14}O_6$	
glucose and Fructose	glucose: $H-C=O$, $(CHOH)_4$, CH_2OH — Aldehyde Fructose: CH_2OH, $C=O$, $(CHOH)_3$, CH_2OH — Ketone	$C_6H_{12}O_6$	
Sucrose		$C_{12}H_{22}O_{11}$	
Diethyl ether	$C_2H_5-O-C_2H_5$	$C_4H_{10}O$	$2C_2H_5OH \xrightarrow[140^\circ C]{H_2SO_4} C_2H_5-O-C_2H_5 + HOH$ C_2H_5OH C_2H_5OH
IMPORTANT IUPAC - propanone Acetone Ketone	$CH_3-C(=O)-CH_3$	C_3H_6O	$CH_3-CH(OH)-CH_3$ (Secondary Alcohol) $\xrightarrow[\text{oxidation}]{[O]} CH_3-C(=O)-CH_3$ (Ketone) $CH_3-C(OH)(OH)-CH_3$ -HOH
Acetaldehyde IUPAC: Ethanal	CH_3CHO or $CH_3C(=O)-H$	C_2H_4O	① From ethyne $HC\equiv CH + HOH \longrightarrow CH_3CHO$ ethyne $H_2C=CH(OH)$ vinyl alcohol — rearrange ② From ethanol CH_3-CH_2-OH (ethanol) $\xrightarrow[\text{oxidation}]{[O]} CH_3C(=O)-H$ $CH_3-CH(OH)(OH)$ — remove HOH
IMPORTANT Vinyl alcohol	$H_2C=CH-OH$ — Vinyl	C_2H_4O	$HC\equiv CH + HOH \longrightarrow H_2C=CH(OH) \xrightarrow{\text{rearrange}} H_3C-C(=O)H$ Vinyl alcohol
Acetic acid	CH_3COOH	$C_2H_4O_2$	$C_2H_5OH \xrightarrow[\text{oxidation}]{[O]} CH_3CHO \xrightarrow[\text{oxidation}]{[O]} CH_3COOH$

COMPOUND NAME	STRUCTURAL FORMULA	MOLECULAR FORMULA	PREPERATION
Trinitroglycerine	CH_2-O-NO_2 $CH-O-NO_2$ CH_2-O-NO_2	$C_3H_5O_9N_3$	CH_2-OH $CH-OH$ CH_2-OH glycerol $+ 3HO-NO_2 \xrightarrow[H_2SO_4]{conc.}$ CH_2-O-NO_2 $CH-O-NO_2$ CH_2-O-NO_2
Formic acid	$HCOOH$	CH_2O_2	CH_3OH (methanol) $\xrightarrow{[O]}$ $HCHO$ (FormAldehyde) $\xrightarrow{[O]}$ $HCOOH$ (formic acid)
IMPORTANT **Oxalic acid** dicarboxylic acid	$COOH$ $COOH$	**$C_2H_2O_4$** In expiremental exam 2023	H_2C-OH H_2C-OH ethylene glycol $\xrightarrow{[O]}$ CHO CHO $\longrightarrow$ $COOH$ $COOH$ oxalic acid
IMPORTANT **Citric acid** 3 COOH groups 1 OH group	$H-C(H)-COOH$ $HO-C-COOH$ $H-C(H)-COOH$	$C_6H_8O_7$	
IMPORTANT **Lactic acid** 1 COOH group 1 OH group	$CH_3-CH(OH)-COOH$	$C_3H_6O_3$	
glycine Amino Acetic acid	H_2N-CH_2-COOH	$C_2H_5O_2N$	$-NH_2-$ $H-C(H)(H)-COOH \longrightarrow H_2N-CH_2-COOH$
FormAldehyde	$HCHO$	CH_2O	CH_3OH (methanol) $\xrightarrow{[O]}$ $HCHO$ (FormAldehyde)
IMPORTANT **Gamixane** IUPAC: HexaChloro cyclohexane		**$C_6H_6Cl_6$** number of atoms = 18 experimental exam 2023 used as insecticide	

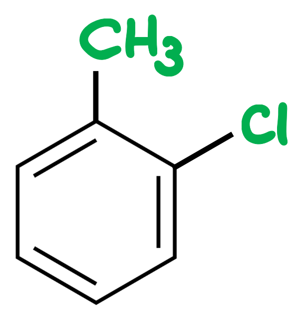
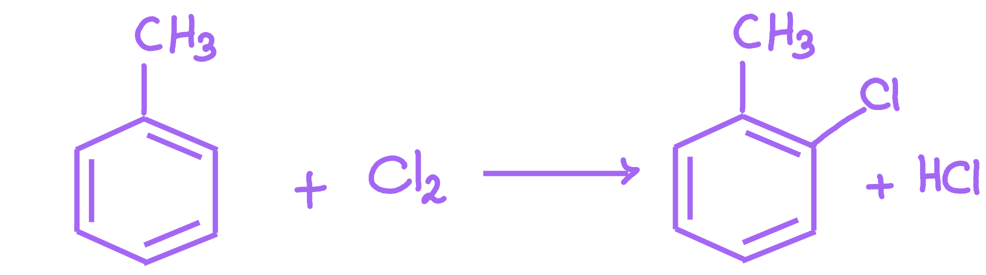
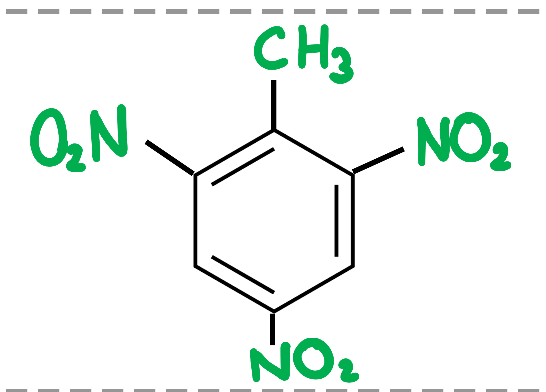
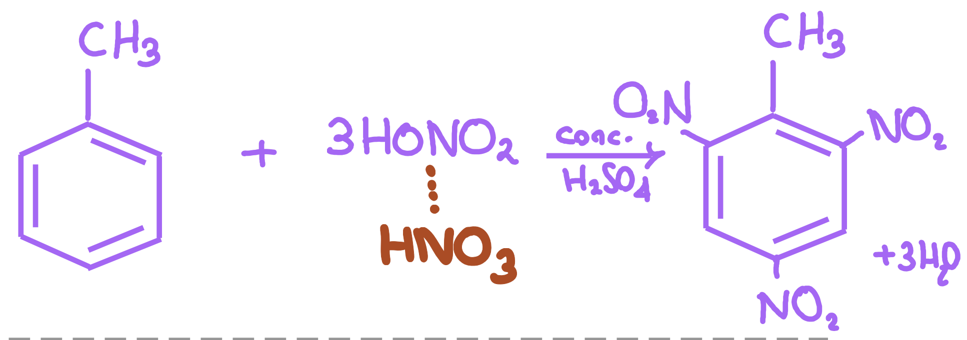
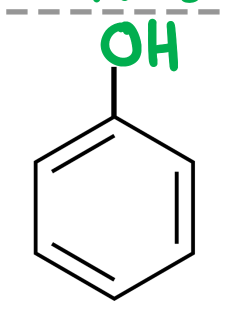
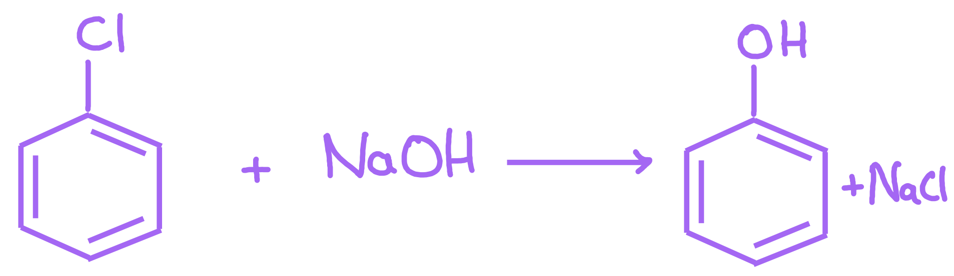
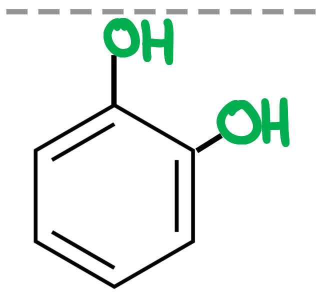
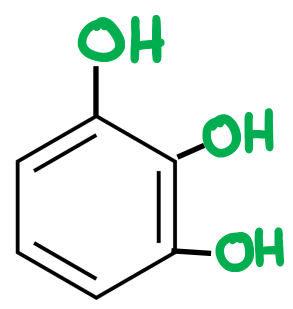
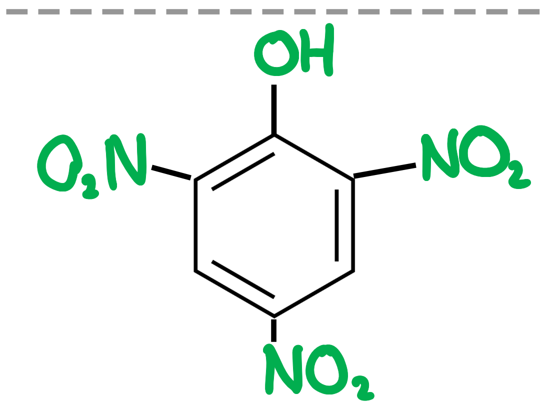
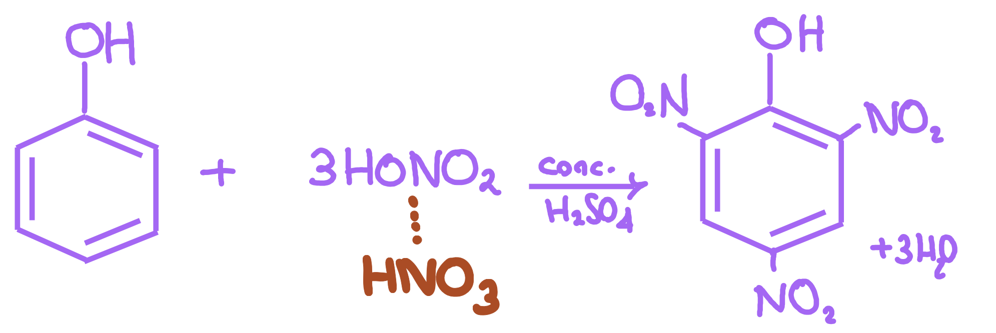
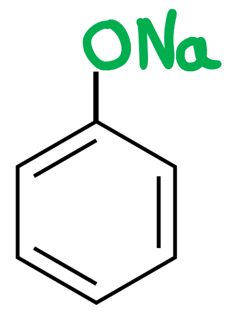
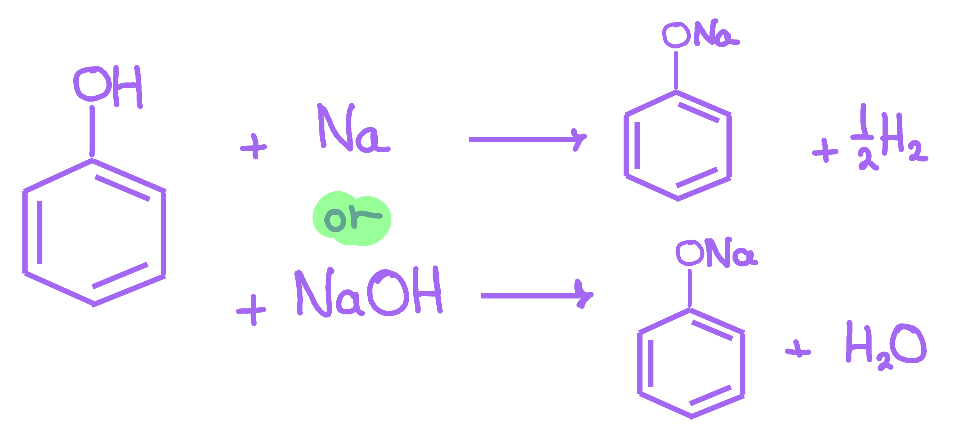
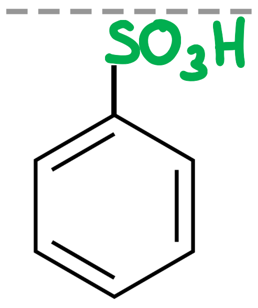
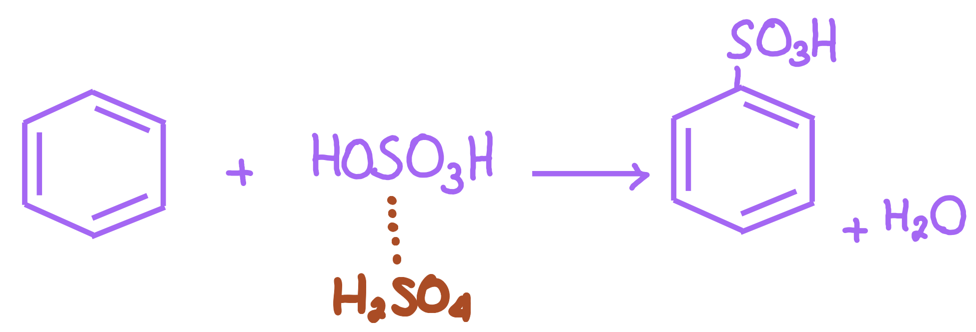
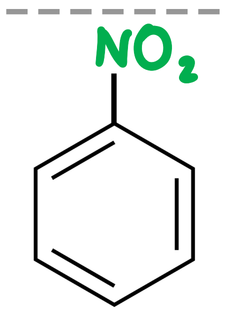
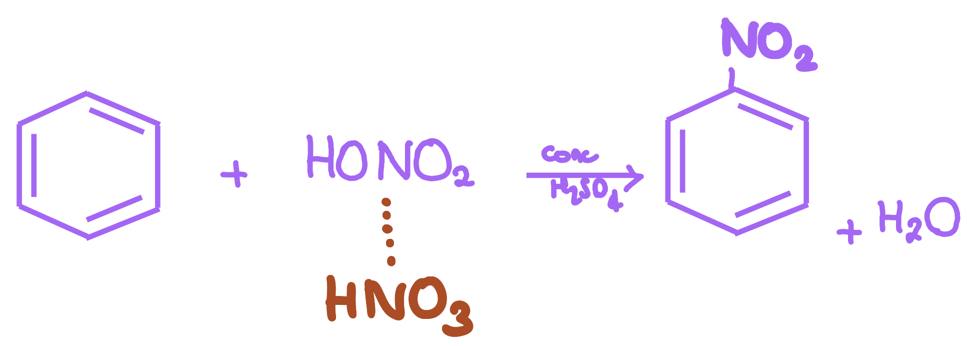
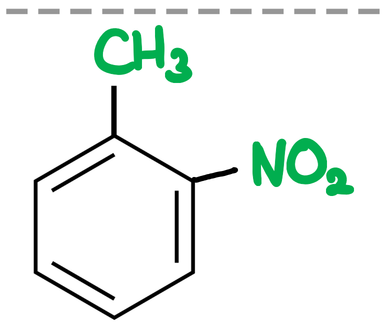
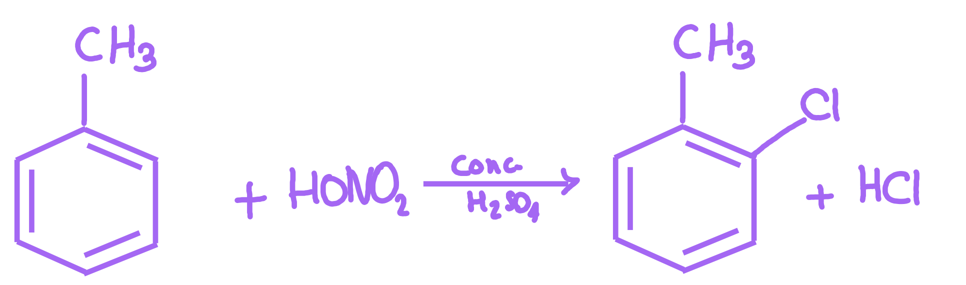
Structural & Molecular Formulas of Important Aromatic Hydrocarbons

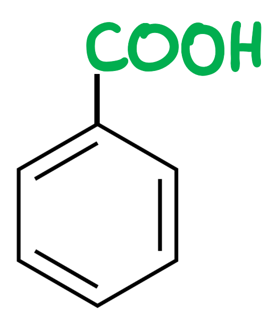
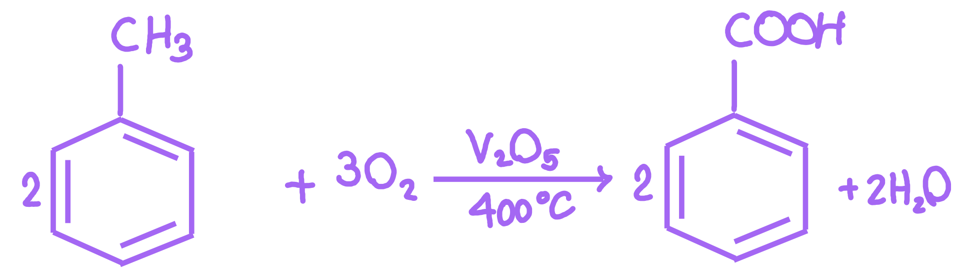
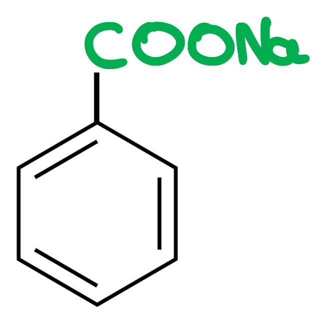
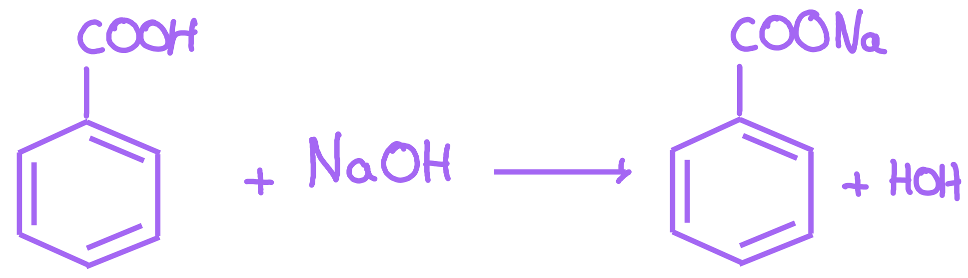
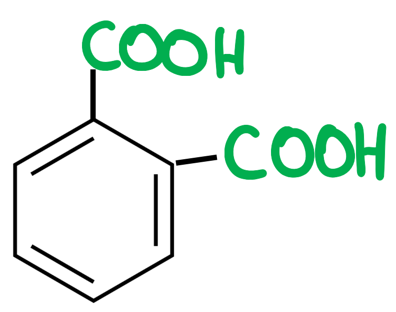
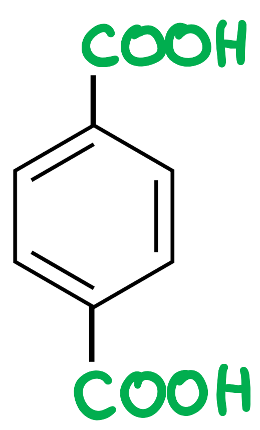
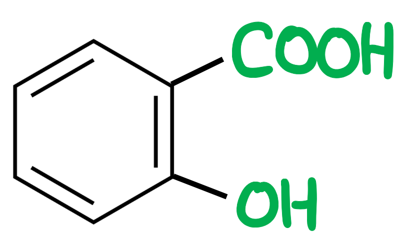
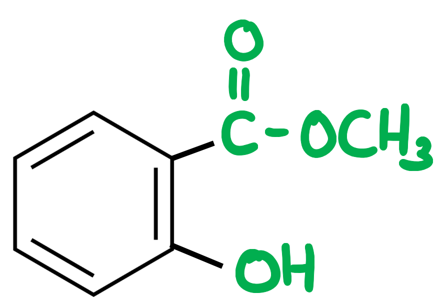
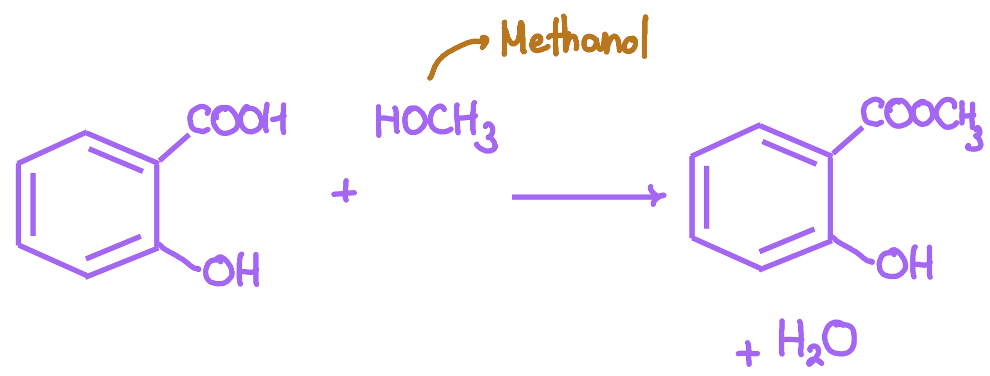
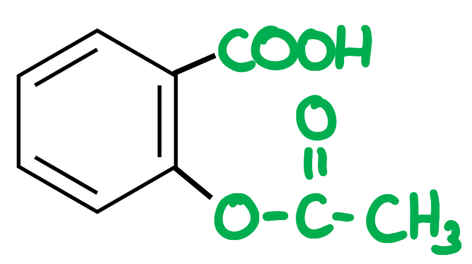
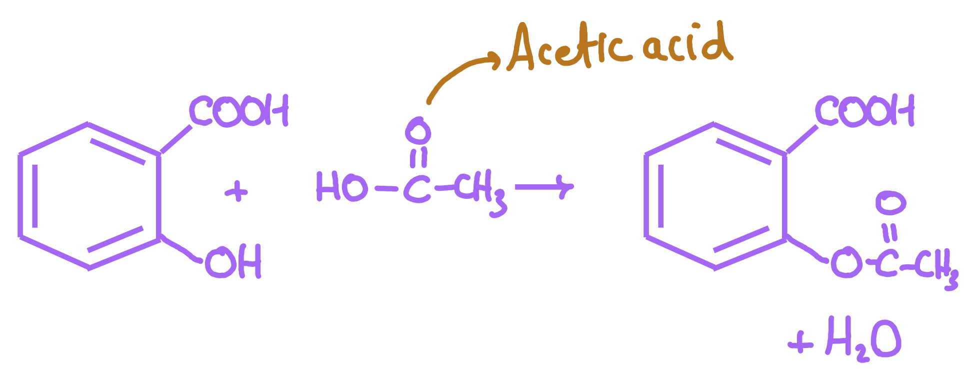
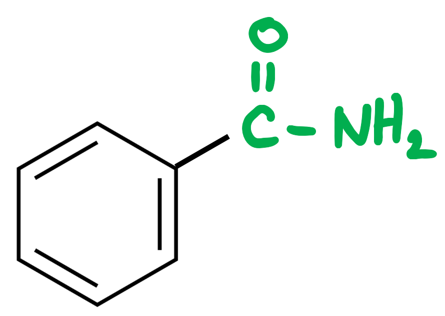
Team Work

COMPOUND NAME	STRUCTURAL FORMULA	MOLECULAR FORMULA	GENERAL FORMULA
1 Aromatic benzene		C_6H_6	C_nH_n IMPORTANT or C_nH_{2n-6} In Experimental Exam 2023
2 Naphtalene		$C_{10}H_8$	-------
3 Anthrathene		$C_{14}H_{10}$	-------
IMPORTANT Toluene		C_7H_8	C_nH_{2n-6}
5 Ethyl benzene		C_8H_{10}	C_nH_{2n-6}
6 1-phenyl, propane IUPAC		C_9H_{12}	C_nH_{2n-6}
7 2-phenyl, propane IUPAC		C_9H_{12}	C_nH_{2n-6}
8 DiphenyL		$C_{12}H_{10}$	

Structural & Molecular Formulas of Important Aromatic Hydrocarbons Drevatives

Team Work

COMPOUND NAME	STRUCTURAL FORMULA	MOLECULAR FORMULA	PREPERATION
ortho chloro toluene		C_7H_7Cl	
IMPORTANT TNT Tri nitro toluene		$C_7H_5N_3O_6$	
phenol Carbolic acid		C_6H_6O C_nH_nO In experimental exam 2023	
Catechol		$C_6H_6O_2$ In expiremental exam 2023	
pyrogalol		$C_6H_6O_3$	
IMPORTANT picric acid Tri nitro phenol		$C_6H_3N_3O_7$	
Sodium phenoxide		C_6H_5ONa	
Benzene sulphonic acid		$C_6H_6O_3S$	
Nitro benzene		$C_6H_5NO_2$	
ortho nitro toluene		$C_7H_7NO_2$	

COMPOUND NAME	STRUCTURAL FORMUlA	MOlECULAR FORMULA	PREPERATION
benzoic acid		$C_7H_6O_2$	
Sodium benzoate		$C_7H_5O_2Na$	
phtalic acid		$C_8H_6O_4$ In experimental exam 2023	
Ter phtalic acid		$C_8H_6O_4$	
IMPORTANT Salicylic acid		$C_7H_6O_3$	
IMPORTANT oil of winter green Marookh oil methyl salicylate ester		$C_8H_8O_3$	
IMPORTANT Aspirin acetyl salicylic acid		$C_9H_9O_4$	
Benzamide		C_7H_7NO	

IMPORTANT

$KMnO_4$ potassium permenganate → Colour changes From violet to Colourless

$K_2Cr_2O_7$ potassium dichromate → Colour changes From orange to green

Team Work

How to Differentiate Between Organic Compounds

BY USING	AlKANE C_nH_{2n+2}	AlKENE C_nH_{2n}	AlKYNE C_nH_{2n-2}
red orange bromine (Br_2) dissolved in Carbon tetra Chloride (CCl_4)	Colour is not removed	Colour is removed	Colour is removed
Adding $KMnO_4$ in alkaline medium	Violet colour isnot removed	violet colour is removed	--

BY USING	AlCOHOL −OH	ETHER R-O-R
piece of Na	reaction occurs and H_2 gas evolves burn with pop Sound	no reaction
Violet $KMnO_4$ in Alkaline medium	violet colour is removed Except For tertiary Alcohol	Violet colour isnot removed
Acidified $K_2Cr_2O_7$	Colour is removed Except For tertiary Alcohol	Colour is not removed

IMPORTANT

BY USING	AlCOHOLS	PHENOlS	ACIDS	ESTERS
Adding Acidified $KMnO_4$	violet colour is removed Except tertiary Alcohols	Violet colour isnot removed	Violet colour isnot removed	Violet colour isnot removed
Adding Acidified $K_2Cr_2O_7$	Colour is removed Except tertiary Alcohol	Colour is not removed	Colour is not removed	Colour is not removed
Na sodium	✓	✓	✓	----
NaOH	✗	✓	✓	✓
HCl	✓	✗	✗	----
Bromine water	---	White ppt. Formed	---	---
Iron III chloride $FeCl_3$	---	Violet Colour Appears	---	---
Na_2CO_3	✗	✗	✓ effervecense	---
$NaHCO_3$	✗	✗	✓ effervecence	---

BY USING	AMMONIA SOLUTION NH_4OH	PHENOL or CARBOLIC ACID	NH_4SCN AMMONIUM THIOCYANAT
Iron III chloride $FeCl_3$	Reddish brown ppt - Formed $FeCl_3 + 3NH_4OH \longrightarrow 3NH_4Cl + Fe(OH)_3$ From Chapter 2	Violet Colour Appears	Blood red colour $FeCl_3 + 3NH_4SCN \longrightarrow 3NH_4Cl + Fe(SCN)_3$ From Chapter 3

Note on Alcohols

1-propanol 1-butanol 1-pentanol etc. Primary 1- اي الكحول فيه

2-propanol 2-butanol 3-pentanol etc. Secondary اي رقم غير ال 1 فيه

2-methyl, 2-propanol 3-methyl, 3-hexanol لو رقم ال Branch وال OH group اتكرر فيه كده Tertiary

Note on Esters

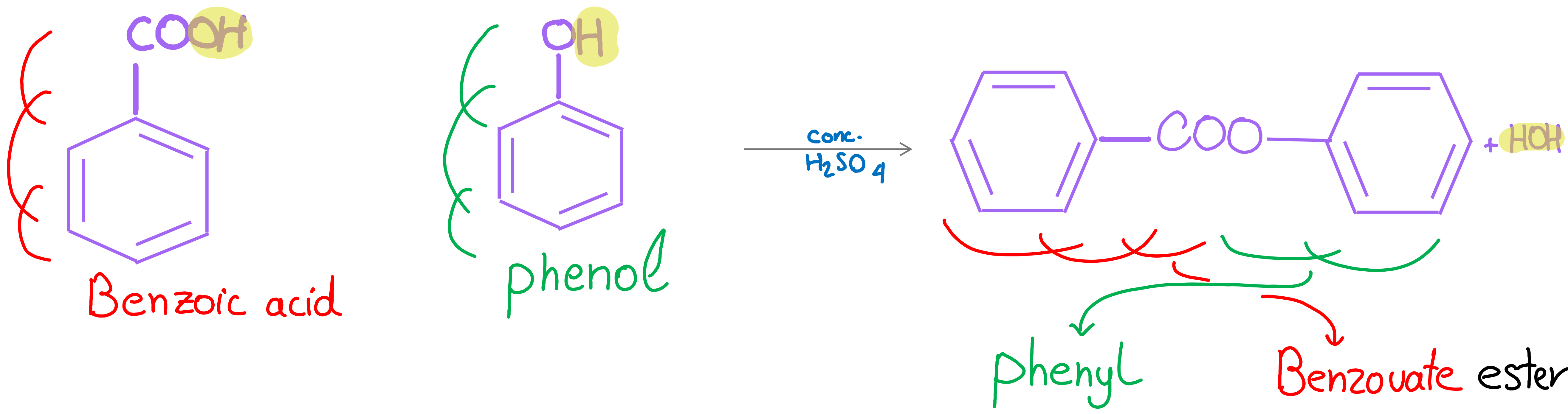
Preparation of esters

$$\text{Acid} + \text{Alcohol} \xrightarrow[H_2SO_4]{conc.} \text{ester} + HOH$$

$$\underbrace{CH_3COOH}_{\text{Acetic acid}} + \underbrace{HOC_2H_5}_{\text{ethanol}} \xrightarrow[H_2SO_4]{conc.} CH_3COOC_2H_5 + HOH$$

ethanol: ethyl alcohol

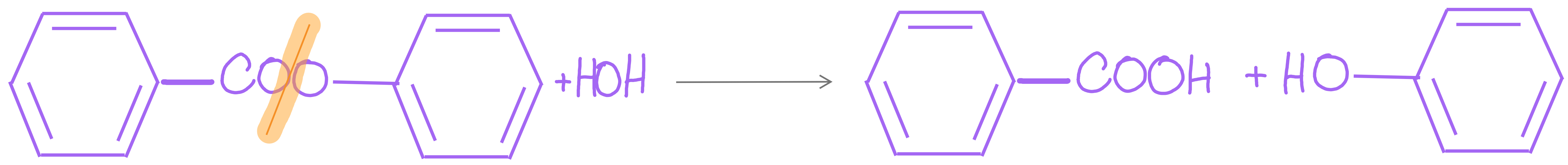
$CH_3COOC_2H_5$: ethyl acetate ester



في اي نوع ال ester hydrolysis
دايما هيتكون Alcohol or phenol

Acidic hydrolysis

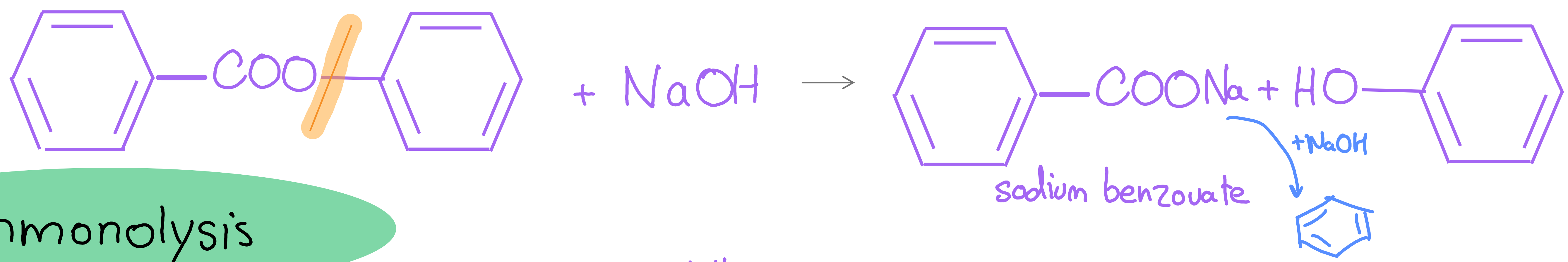
$$CH_3CO\cancel{O}C_2H_5 + HOH \longrightarrow CH_3COOH + HOC_2H_5$$



Alkaline hydrolysis

$$CH_3CO\cancel{O}C_2H_5 + NaOH \longrightarrow \underbrace{CH_3COONa}_{\text{Sodium Acetate}} + HOC_2H_5$$

$$CH_3COONa \xrightarrow{+NaOH} CH_4 \text{ methane}$$



Ammonolysis

$$CH_3CO\cancel{O}C_2H_5 + \overset{NH_3}{HNH_2} \longrightarrow \underbrace{CH_3CONH_2}_{\text{Acet amide}} + HOC_2H_5$$